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RESEARCH ARTICLE



Beyond the geotag: assessing implicit geoprivacy risks in visual user-generated content

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ABSTRACT

User-Generated Content (UGC) on social media has created a wealth of geographic information. However, it also presents a significant geoprivacy challenge. Image geolocation is the task of identifying an image's location. It offers benefits in domains such as disaster response and autonomous driving. Concurrently, it enables the inference of personal locations from implicit geographic cues within the images. Thus, quantitatively assessing the implicit geoprivacy risk of images is becoming increasingly important. To address this problem, this study constructs a Bayesian Network that includes (1) a comprehensive feature set of 11 metrics in four categories of an image's geolocation, (2) a taxonomy of five core geolocation methods, modeling the causal pathways from the features to the methods, and finally to the vulnerability. We constructed a dataset with 42,576 real-world examples of image geolocation. The model's AUC reaches 0.90 on the test dataset (where 1.0 is perfect and 0.5 is random chance). The results also support that task attractiveness, scene complexity, and landmark level are vital factors in determining the final localization probability. This study provides granular insights into how specific features influence the choice of localization pathways.

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Geoprivacy; geo-location; risk assessment; bayesian network; volunteered geographic information

1. Introduction

The proliferation of social media and pervasive mobile technologies fundamentally reshaped the interactions between individuals and the physical world. As such, User-Generated Content (UGC) increases substantially with explicit or implicit geographic information. The convergence of GIS and social media, a trend previously identified by scholars such as Sui and Goodchild (2011), created unprecedented opportunities across numerous domains, including urban planning (Liu *et al.* 2017), disaster response (Yin *et al.* 2026), and the study of human mobility (Chen *et al.* 2019). However, these advancements also introduced critical challenges related to geolocation exposure, defined as the unauthorized disclosure of an individual's location and movements (Leitner and Curtis 2006).

Early studies on geoprivacy primarily focused on explicit geographic data, such as GNSS coordinates (Leitner and Curtis 2006, Rao *et al.* 2021) or personal trajectory data (Rao *et al.* 2023). However, a more subtle and pervasive threat emerged from implicit geographic cues embedded within social media content, including images. When exploited maliciously, image geolocation technologies pose a significant threat to privacy. Publicly shared images on social media can inadvertently reveal precise location information and personal activity patterns (Morris *et al.* 2023), which may in turn expose other sensitive information and heighten an individual's vulnerability to stalking, identity theft, and physical harm (Cohen-Almagor and Trottier 2022).

Over the past decade, a growing body of research has sought to assess and mitigate risks associated with social media images (Ghazinour and Ponchak 2017, DeHart *et al.* 2020, Hasan 2020). Some recent efforts turned to Large Language Models (LLMs) to address location disclosure risks and employed these models both to identify and reduce privacy threats through techniques such as image obfuscation (Hasan *et al.* 2018, Li *et al.* 2020, Ma *et al.* 2025). These studies demonstrate that the combination of various implicit geographic cues can substantially increase the risk of geolocation exposure. Nevertheless, existing research still lacks a quantitative framework for evaluating a photo's vulnerability to location disclosure and for deconstructing the geolocation inference process. This gap hinders the public's ability to identify images that are highly susceptible to geolocation.

Furthermore, the integration of Large Vision-Language Models (VLLMs) into geolocation tasks introduces profound ethical challenges regarding the unauthorized surveillance of individuals. As Geospatial AI (GeoAI) lowers the technical barrier for extracting sensitive location data from unstructured media, the ethical imperative shifts from merely utilizing spatial data to governing its potential for misuse. Consequently, establishing a risk assessment framework is not only a technical necessity but also a fundamental step towards ensuring accountability and protecting user consent in the era of generative AI.

Consequently, this study aims to develop a comprehensive framework for assessing geoprivacy risks. Specifically, the proposed approach seeks to achieve three objectives: first, to quantitatively evaluate the inherent geoprivacy risk of a photograph; second, to identify the critical visual and contextual geographic cues that contribute to this risk; and third, to analyze potential inference pathways (i.e., localization methods) facilitated by these cues.

The remainder of this paper is structured as follows. [Section 2](#) provides a literature review on image geolocation, covering key features and methods. [Section 3](#) elaborates on the construction of a Bayesian Network-based vulnerability assessment model. [Section 4](#) details the experimental design and presents and discusses the results. Finally, [Section 5](#) examines the implications and limitations of this study and suggests directions for future research.

2. Related works

The primary objective of this study is to develop a robust model for the quantitative assessment of image disclosure risk. The first step is to understand the mechanisms

underlying location inference. Accordingly, the literature review is structured around two core components: the specific features of user-generated content (UGC) that function as geographic cues and the methods that utilize these features to geolocate images.

2.1. Image geolocation features

In the field of computer vision (CV), a recent and relevant classification of image geolocation features was proposed by Jia *et al.* (2024), who categorized intrinsic image features into continuous and discrete geographical features.

Continuous geographical features refer to visual elements that vary smoothly and gradually across space, exhibiting strong local correlation between visual appearance and geographic coordinates. For example, the transition in vegetation from temperate forest to alpine meadow or the subtle shift in architectural styles across a continent are continuous features that a model can learn to associate with broad geographical regions. This category is well-established in CV, particularly for worldwide image geolocation. The underlying principle involves training deep learning models on large datasets of image-GNSS pairs to predict the location of a given photograph. Current approaches to this task mainly fall into two categories: classification-based and retrieval-based methods.

Classification-based methods, pioneered by models such as Planet (Weyand *et al.* 2016), discretize the globe into a grid of cells and learn common visual characteristics within each cell, thereby capturing implicit geographical continuity. Subsequent research kept focusing on enhancing this property. For example, multi-scale hierarchical approaches (Müller-Budack *et al.* 2018, Clark *et al.* 2023) explicitly modeled continuity through a layered structure. In contrast, state-of-the-art models such as PIGEON (Haas *et al.* 2024) employed distance-aware loss functions to directly enforce spatial continuity in the learned features.

Retrieval-based methods (Lee *et al.* 2022, Vivanco Cepeda *et al.* 2023, Shao *et al.* 2023), on the other hand, directly model geographical continuity by using contrastive learning to map images and their corresponding GNSS coordinates into a shared embedding space. This approach inherently ensured that visual features capture geographical continuity, as proximity in the vector space corresponds to real-world proximity. Recent advancements in this area integrated Retrieval-Augmented Generation (RAG) to further enhance the model's perception of global geographical continuity, with experimental results demonstrating satisfactory performance (Jia *et al.* 2024, Zhou *et al.* 2024).

Discrete geographical features, in contrast, exhibit distinct and abrupt visual characteristics across space. These features are best understood as structural elements whose geometric shapes and spatial arrangements are discernible from both ground-level and overhead (e.g., satellite) perspectives. Unlike continuous features that provide broad regional context, discrete features offer precise locational anchors through cross-view matching. Cross-View Geo-Localization (CVGL) has been shown to be effective in this context (Durgam *et al.* 2024). CVGL operates by matching a ground-level query image against a database of high-resolution satellite imagery to

retrieve precise coordinates. Unlike global-scale methods that model feature continuity across large distances, CVGL leverages the unique spatial arrangement of discrete elements—such as building footprints, road layouts, and specific landmarks—within a constrained local area. As a result, CVGL achieves strong performance on various urban benchmarks (Huang *et al.* 2025) and underscores that a focus on discrete geographical features is key to achieving high precision in fine-grained, small-area localization.

However, while foundational, the continuous-discrete classification of features cannot fully encompass all the elements utilized by analysts in a UGC context (where ‘analysts’ refers to any individuals attempting to locate an image, ranging from Open-Source Intelligence (OSINT)¹ practitioners to skilled hobbyists and other curious online users). In addition to intrinsic image features, off-image features and image quality also play critical roles in geolocation.

Off-image features refer to extrinsic data associated with an image’s publication context. The features—which include publisher metadata (e.g., the user’s profile description, post history, or social network connections) and explicit textual clues (e.g., the image title, caption, user comments, or hashtags)—provide rich spatial and social contextual information. Although image geolocation is well-established in domains such as OSINT (Ivanjko and Dokman 2019, Yadav *et al.* 2023), much of the computer vision literature remains focused primarily on pixel-level data, often overlooking these contextual cues. The predictive power of non-visual data is well recognized within GIScience. For example, Pellet *et al.* (2019) demonstrated that a user’s location could be accurately inferred through social network connections and textual content alone. Similarly, studies on human mobility prediction successfully incorporated factors such as temporal patterns and historical visits within Bayesian models to forecast future locations (Chen *et al.* 2019), which further underscored the importance of extra-visual spatio-temporal context.

Image quality for geolocation encompasses technical properties that affect the fidelity and interpretability of visual content. Attributes, such as weather conditions, viewing angle, clarity, focal length, and lighting, were shown to significantly impact geolocation accuracy (Durgam *et al.* 2024, Haas *et al.* 2024). The importance of these factors is evident in research on orientation modeling. For instance, Liu and Li (2019) and Shi *et al.* (2020) demonstrated that explicitly modeling viewpoint and camera orientation could substantially improve model performance. Furthermore, models such as PIGEON (Haas *et al.* 2024) achieved state-of-the-art results by augmenting training data with geographical, climatic, and orientation metadata, which affirms that incorporating quality variations is essential for effective image geolocation.

2.2. Image geolocation methods

The process of image geolocation generally follows a hierarchical ‘coarse-to-fine’ trajectory, which progresses from broad regional identification to fine-grained pinpointing of an exact location. As observed by Mehta *et al.* (2016), this multi-stage strategy is fundamental to addressing complex geolocalization challenges.

1. **Coarse-Grained Localization.** The initial stage aims to narrow the search space to a plausible country or region. This approach is demonstrated in the popular geolocation game GeoGuessr, where players must deduce their coarse location from visual cues within interactive panoramic imagery. As noted by Mehta *et al.* (2016), the experts often leverage experiential knowledge to integrate subtle visual cues (e.g., the type of soil and rock formations, the style of electrical outlets or power lines, the color and format of license plates) into a coherent geographical hypothesis. In addition, non-expert methods can also be effective; for example, Kim *et al.* (2022) demonstrated that crowdsourced consensus can achieve reliable coarse localization. Moreover, in a UGC context, methods may bypass visual analysis entirely. Pellet *et al.* (2019), for instance, showed that social network connections and textual metadata can be used to infer a user's probable location. Overall, a common outcome of this stage is the identification of a broad area of interest rather than precise coordinates.
2. **Fine-Grained Localization.** Once a general region is established, fine-grained localization techniques are employed to identify the exact location. In some cases, this step can be accelerated or bypassed using search engine-based methods. Recent research (Wazzan *et al.* 2024) indicates that big-data-enhanced search techniques are effective for geolocation—in certain contexts surpassing state-of-the-art VLLM-based reasoning. When direct matching is not feasible, a subsequent step often involves spatial modeling. This entails two tasks: analysts first construct a mental or digital model of the scene's spatial layout, then systematically scan satellite or street-view imagery of the target region to identify a matching configuration.

In summary, existing research has made considerable progress in identifying geographic cues and developing methods for image geolocation. Nevertheless, two critical research gaps remain. First, from a feature perspective, current studies have largely focused on extracting visual cues while neglecting to quantify their relative importance in the geolocation inference process. Moreover, most studies consider only a limited set of features, overlooking the essential roles of contextual off-image features—especially in a UGC context—and image quality within geolocation frameworks such as geoprivacy risk assessment. Second, from a methodological standpoint, numerous strategies have been proposed for geolocation, and the effectiveness of these methods, accessible to both experts and non-experts, underscores a critical reality: nearly any piece of publicly available user-generated content (UGC), including images and videos, can potentially be geolocated. This emphasizes the urgent need for a systematic framework to assess the probability and mechanisms of implicit location disclosure.

3. Methods

The proposed vulnerability analysis of image location disclosure is constructed on a corresponding indicator framework. The framework, comprising three hierarchical levels, models the logical progression of a real-world geolocation analysis: features, methods, and vulnerability.

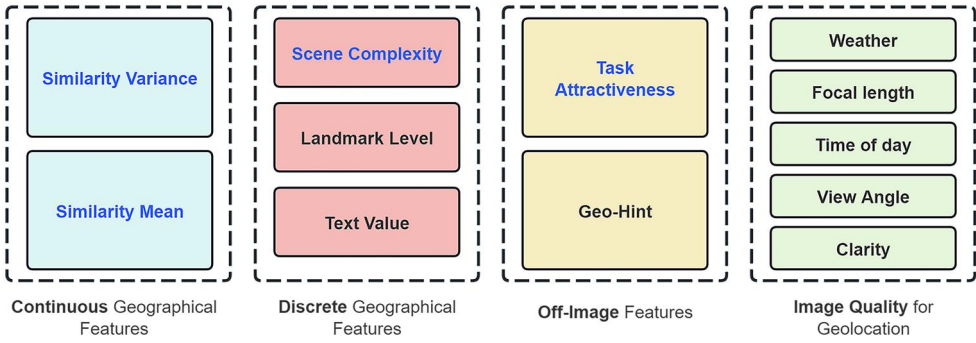


Figure 1. Classification of image geolocation features.

Level 1, Features (The Evidence). This foundational level comprises all observable evidence, both intrinsic and extrinsic to the image.

Level 2, Methods (The Process). This intermediate level represents the various cognitive strategies or technical methods employed by analysts. The choice and efficacy of these methods are directly influenced by the available features.

Level 3, Vulnerability (The Outcome). This final level represents the outcome—the probability of successful geolocation resulting from the methods applied.

3.1. Modeling process for image geolocation features

Building on existing research, this paper categorizes geolocation features into four groups: continuous geographical features, discrete geographical features, Off-Image Features, and Image Quality for Geolocation.

Figure 1 presents the complete feature architecture. Blue-annotated features are newly designed to abstract and quantify relevant underlying characteristics, while other features are analyzed by the VLLM using structured prompts.

This study employs Qwen-VL-Max VLLM (Bai *et al.* 2023) for its robust visual analysis capabilities, multilingual Optical Character Recognition (OCR) proficiency, and its demonstrated ability to follow complex instructions. While the accuracy of these semantic features depends on the model's capabilities, the overall framework is designed to be model-agnostic, allowing for future integration of more advanced VLLMs. Appendix A provides illustrative examples of the prompts used.

3.1.1. Quantitative analysis of continuous geographical features

To quantitatively assess the contribution of continuous geographical features in the image geolocation task, we proposed the Geo-Uniqueness Score. Geo-Uniqueness comprises two components: Similarity Variance and Similarity Mean. The calculation of these components involves two steps: (1) the construction of a Global Reference Geographical Feature Library (Refer-Library), and (2) the subsequent calculation of the Geo-Uniqueness metrics.

Both steps utilized GeoClip (Vivanco Cepeda *et al.* 2023), a CLIP-based model trained on extensive image-GNSS coordinate pairs. We selected GeoClip as our feature

extractor due to its efficacy in identifying geographical features within visual data. Additionally, its training dataset serves as the foundation for our Refer-Library.

1. Construction of a global reference geographical feature library (Refer-Library)

Constructing the Refer-Library requires a sampling strategy that balances global coverage with real-world relevance and feature quality. A purely uniform sampling method is suboptimal, as it would be dominated by vast, homogeneous regions (e.g., open oceans) that offer minimal discriminative information. Therefore, we employed a two-step ‘generate-then-filter’ strategy.

We first generated a large set of candidate GNSS coordinates, sampled globally using the Equal Earth projection (Šavrič *et al.* 2019) to ensure comprehensive initial coverage. Subsequently, these candidates are filtered by retaining only those located within a predefined spherical distance (e.g., 100 km) of any data point in the GeoClip training dataset. Comprising millions of geotagged images, this dataset serves as a proxy for frequently photographed locations globally.

This filtering step is crucial for two reasons. First, the filtering step concentrates the reference points in regions relevant to User-Generated Content (UGC), effectively excluding vast, rarely photographed areas such as open oceans or deserts. Second, the filtering step enhances the reliability of the feature library, as GeoClip’s encoder produces reliable embeddings for locations similar to those in its training data. This filtering step enables the final set of N GNSS anchor points to maintain global coverage while ensuring the feature quality of each sample.

Finally, these N anchor points are batch-encoded using GeoClip’s pre-trained location encoder. This process transforms each GNSS coordinate into a D -dimensional, L2-normalized geographical feature vector, which collectively forms the Refer-Library V_{gps} . We set the dimension to $D=512$, aligning with GeoClip’s native output to balance feature richness with computational efficiency.

2. Calculation of geo-uniqueness

For a given input image I_q , we employ the GeoClip’s Image Encoder $E_{\text{img}}(\cdot)$ to extract its visual feature vector $v_{\text{img}} = E_{\text{img}}(I_q)$. Subsequently, we compute the cosine similarity s_j . This metric measures the visual alignment between the input image and a specific coordinate. A score close to 1.0 indicates that the visual content of the image (e.g., its architecture, landscape, and textures) highly correlates with that coordinate. Figure 2 illustrates this process, comparing two distinct query images against a random sample of 2,000 reference images extracted from the CVUSA dataset (Workman *et al.* 2015).

This similarity is calculated between the image feature vector and each geographical feature vector $v_{\text{gps}}^{(j)}$ in the Refer-Library V_{gps} .

$$s_j = \text{sim}\left(v_{\text{img}}, v_{\text{gps}}^{(j)}\right) = \frac{v_{\text{img}} \cdot v_{\text{gps}}^{(j)}}{\|v_{\text{img}}\|_2 \cdot \|v_{\text{gps}}^{(j)}\|_2} \quad (1)$$

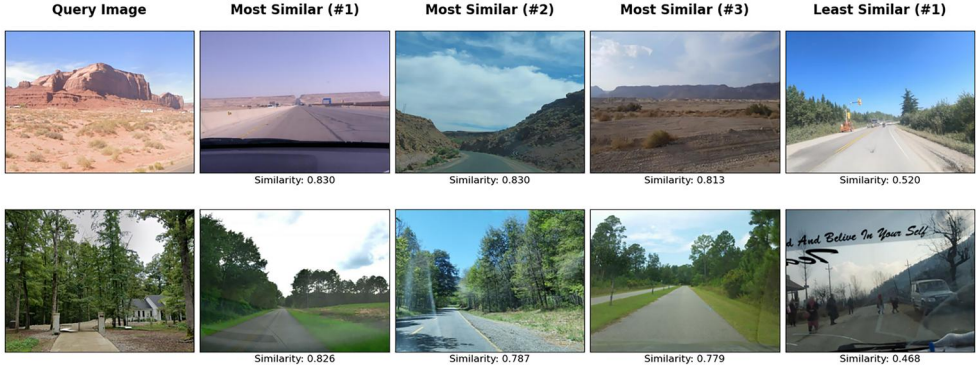


Figure 2. Visualization of the continuous geographical feature extraction process. The leftmost column displays query images, followed by their three most similar matches and the single least similar match from the reference library.

This process yields a set of N similarity values, $S = \{s_1, s_2, \dots, s_N\}$. From this set, two statistical metrics are calculated to jointly characterize the continuous geographical features of the image:

Similarity Mean ($\bar{s}$): This metric denotes the arithmetic mean of similarity score across all anchor points.

Similarity Variance ($\text{Var}(S)$): Defined in Equation (2), this metric measures the dispersion of the image's similarity scores across all global anchor points.

$$\text{Var}(S) = \frac{1}{N-1} \sum_{j=1}^N (s_j - \bar{s})^2 \quad (2)$$

We independently normalize both Similarity Variance and Similarity Mean across the dataset via min-max scaling. Subsequently, we discretized these normalized values into K intervals through quantile-based binning. In this study, $K=10$ was chosen as a pragmatic parameter to balance the trade-off between information granularity and the complexity of the network's Conditional Probability Tables (CPT).

The resulting discrete states for Similarity Variance and Similarity Mean are then used as two separate evidence nodes in the Bayesian Network. This dual-indicator approach captures the statistical profile of an image's continuous geographical features, allowing the model to learn the distinct patterns of 'unique' (high variance, low mean), 'generic' (low variance, high mean), and 'uninformative' (low variance, low mean) scenes.

3.1.2. Quantitative analysis of discrete geographical features

Discrete geographical features are defined by spatially distinct and discontinuous visual characteristics. These features often lack a significant linear correlation with geographic coordinates but play a crucial role in the localization process. In this study, we categorize these features into text value for geolocation (e.g., numbers and signage), landmark level (including their prominence), and scene complexity (Jia *et al.* 2024).

We employ the previously introduced Qwen-VL-Max model to quantify the text value and landmark level. For the analysis of Scene Complexity, we define

Environmental Structural Complexity (ESC) as the quantity and diversity of geolocation-relevant visual elements within an image.

The calculation of ESC begins with the identification of visual elements using a pre-trained semantic segmentation model, HRNetV2 (Wang *et al.* 2021), implemented via the MMSegmentation toolbox (MMSegmentation Contributors 2020). From the full set of segmented objects, we select a subset of categories relevant to geolocation (e.g., buildings, roads, and specific landforms) as the scene elements because these categories constitute the structural backbone of the scene. Appendix B details the complete list of the selected visual categories.

Finally, we quantify the scene complexity by calculating the total number of distinct object instances within this filtered subset. We first normalize this raw count across the entire dataset via min-max scaling, and subsequently discretize the continuous values into K intervals using equal-width binning. This process yields the final scene complexity score for the Bayesian network.

3.1.3. Quantitative analysis of off-image features

3.1.3.1. Task attractiveness score. The potential for an image to be successfully geolocated depends not only on its visual content but also on an external factor: its task attractiveness. We propose this metric to quantify the likelihood that a specific image motivates a geolocation attempt. A task with high attractiveness does not correlate with location sensitivity, but rather that the social context of the post makes it a more compelling target for analysis. We model task attractiveness through two key components: the author influence factor (AIF) and content appeal.

Social media research establishes publisher influence as a strong predictor of content visibility and engagement (Suh *et al.* 2010). Existing studies quantify publisher influence via composite metrics that combine user reach (e.g., follower count) and engagement levels (Cha *et al.* 2010). Adapting this principle to the UGC context, we proposed the Author Influence Index (All). Content from influential publishers naturally attracts more attention, thereby increasing the probability of a geolocation attempt. We calculate All as follows:

$$All_{log} = w_u \cdot \log(1 + u) + w_f \cdot \log(1 + f) \quad (3)$$

where u denotes the number of user followers, f indicates the average number of likes per post, and w_u and w_f are the corresponding weight coefficients. For the current model configuration, we set $w_u = 5.0$ and $w_f = 1.0$. Subsequently, we apply Min-Max normalization to the raw All to derive the final Author Influence Factor (AIF).

In addition to author influence, the content appeal of a post also drives user attention. We modeled this factor through two proxies: Timeliness, measured as the elapsed duration since the post's last interaction, and Historical Context, represented by a binary variable (IsAncient) indicating if community tags deem the post notably old.

To synthesize these disparate off-image factors into a single, comprehensive metric, we formulate the Task Attractiveness Index (TAI). Grounded in risk assessment, this index treats AIF as a proxy for exposure, and Content Appeal as the post's intrinsic attractiveness. Consequently, TAI quantifies the overall likelihood that an image solicits a geolocation attempt:

$$TAI = \left(\alpha \left(1 - \frac{t_{\text{current}} - t_{\text{comment}}}{t_{\text{current}} - t_{\text{create}}} \right) + (1 - \alpha)AIF \right) \times (1 - \beta \cdot \text{IsAncient}) \quad (4)$$

Where t_{create} , t_{comment} , and t_{current} denote the timestamps of post creation, the last comment, and the data collection cutoff, respectively. The ratio term thus quantifies the post's temporal relevance. AIF is the previously defined Author Influence Factor. The binary variable IsAncient takes the value 1 when the community classifies the post as 'Ancient Problem', and 0 otherwise. We configure the weighting coefficients as $\alpha = 0.6$ and $\beta = 0.8$, empirically prioritizing recency while penalizing historical obsolescence.

We design TAI as a relative indicator for benchmarking the attractiveness of posts within a specific platform context. While the current formulation leverages metadata specific to the ChaoFun platform, the framework remains adaptable. For platforms such as Reddit or X (formerly Twitter), researchers can integrate alternative engagement metrics (e.g., retweets, shares, or saves) into the TAI formulation to accurately capture the dynamics of task attractiveness in those specific ecosystems.

3.1.3.2. Geo-Hint (geographic clue exposure). We proposed the Geo-Hint metric to quantify the exposure of explicit geographical clues embedded in textual metadata. Employing the Qwen-VL-Max model, we analyze all available text associated with an image, such as its title, description, or user comments. The model extracts mentioned geographical entities and classifies them into one of five pre-defined exposure levels based on spatial granularity: none, country-level, province/state-level, city-level, or district/county-level.

3.1.4. Quantitative analysis of image quality for geolocation

Image quality for geolocation encompasses factors that can significantly influence the photo localization process. poor image quality degrades visual information, causing feature mismatches even for photos taken at geographically close locations. While existing geolocation studies view poor image quality as a challenge to be overcome, our framework interprets quality attributes as independent, quantifiable risk variables. This process allows for the quantification of their direct contribution to location disclosure risk. Finally, we identify five key metrics to assess image quality for geolocation. These metrics, along with their possible states, were detailed in Table 1.

Except for clarity, we evaluated all image quality metrics and the Geo-Hint level employing the Qwen-VL-Max model. For clarity, we computed the Variance of

Table 1. Metrics related to image quality for geolocation.

Feature name	Possible states	Notes
Weather	Other, Cloudy, Sunny, Overcast, Rainy, Snowy, Foggy	'Else' indicates indoor scenes or indescribable weather conditions.
Time of day	Night, Daytime, Dusk, Dawn, Indoor	'Dusk' and 'Dawn' are visually similar; 'None' indicates indoor scenes.
View angle	Low-angle, Eye-level, Aerial-level, High-level	None
Focal length	Average, Long, Wide	Indicates the photo's field of view (FOV).
Clarity	State0—State9	Discretized and normalized variance of the Laplacian.

Laplacian (VOL) (Nayar and Nakagawa 1994). We normalized these values via min–max scaling and subsequently discretized them into 10 distinct clarity intervals.

3.2. Modeling process for image geolocation methods

In the geolocation process, the methods used by analysts are complex and not directly observable. To capture this intermediate reasoning step, we model these geolocation methods as conceptual latent variables (represented by the hidden nodes) in Level 2 of the Bayesian Network. We categorize these methods according to the features they primarily utilize. Based on preliminary investigation, we categorize current image geolocation methods into five types:

1. **The User Background Analysis.** This OSINT-based approach focuses on the individuals who published the image. By profiling the subject's likely geographical footprint, this method constrains the potential search space. Analysts systematically investigate the subject's digital trail, aggregating historical posts, social activities, professional affiliations, and any other publicly available personal information. By correlating these disparate data points, analysts infer movement patterns and identify key locations—such as residence, workplace, or recent travel destinations. For example, analysts might discover through a social media bio that the subject is a student at a specific university. The search can then be focused on the vicinity of that university's campus and the surrounding areas.
2. **The Search Engine-based Method.** This method leverages reverse-image search technology to retrieve online instances of the target image or its constituent elements. By directly sourcing location-specific information from the web, this approach aims to bypass the deductive localization process. This method involves uploading the entire image, or a cropped section containing a distinct object, to various search engines (e.g., Google Images, Yandex, Baidu). The engine's algorithms then match the query against billions of indexed images. A successful hit can lead to a webpage, article, or social media post that explicitly names the location or provides critical context. For example, an image containing a unique sculpture, when reverse-searched, might lead to a travel blog titled 'My Visit to Barcelona', thereby directly disclosing the location.
3. **Environmental Detail Analysis.** This foundational approach relies on the systematic observation and interpretation of visual clues within the image. By refining a geographical hypothesis, this approach constrains the vast global search space to manageable regions. This approach adopts a hierarchical 'funnel' workflow: extracting broad features (e.g., vegetation, architecture), making a regional judgment (e.g., climate zone, country), comparing unique details (e.g., infrastructure, signage), and progressively reducing the scope. Analysts synthesize these observations to construct deductive chains, continuously refining their search area by eliminating inconsistent regions. For example, the conjunction of tropical foliage and left-hand traffic restricts potential locations to Southeast Asia or Oceania, while a specific style of power outlet could pinpoint the exact country.

4. **Spatial Modeling.** This approach facilitates fine-grained localization by exploiting spatial relationships between objects in a scene. Especially effective in the absence of textual or landmark clues, this approach achieves precise coordinates by matching a reconstructed scene geometry with map interfaces (e.g., satellite or street-view cartography). Analysts abstract the scene's layout mentally or with digital tools based on the relative positions and orientations of structural elements (e.g., buildings, roads, and terrain). Subsequently, they systematically scan a map interface of the target area, searching for a configuration that geometrically aligns with their abstracted layout. For example, within a candidate neighborhood, analysts might model the alignment of a church steeple, a crossroads, and a bridge. Then they scan the area on Google Earth to find the one spot that matches this unique spatial signature.
5. **The External Tool Application.** This approach augments visual analysis by querying specialized online databases and utilities. By corroborating geographical hypotheses or yielding definitive solutions, this approach provides precise auxiliary information. Unlike general web searching, this approach requires extracting discrete identifiers—such as aircraft tail numbers, license plates, or storefront typography—to execute targeted queries within domain-specific repositories. Consequently, this method serves a function to transform an ambiguous clue into a concrete data point. For example, analysts might use Flightradar24 to identify a plane visible in an image or consult a business directory like Yelp to identify a restaurant. They can also use a weather archive to confirm the atmospheric conditions on a specific day.

3.3. Construction of a Bayesian network

The effectiveness of Bayesian Networks (BNs) in modeling the complex, multimodal nature of privacy risk perception was demonstrated in prior studies (Alrayes *et al.* 2020, Hu *et al.* 2024). Therefore, we employ BN to model the intricate relationships between features, methods, and the final risk outcome. Figure 3 illustrates our proposed architecture, which delineates the causal relationships between evidence nodes (features), hidden nodes (methods), and the target output node (vulnerability).

To operationalize this structure, we defined the specific probabilistic dependencies by restricting the connections from evidence nodes to their corresponding hidden method nodes. Drawing inspiration from studies of human expert behavior, such as the 'zooming-in' strategy observed by Mehta *et al.* (2016), the localization is classified into a two-stage process: establishing high confidence in a general region and pinpointing the exact location.

To mediate the transition between these two steps, we introduced a node named Region Lock-in Confidence to quantify the certainty of the coarse-grained localization phase. We connect the methods excelling at regional identification to this node, including User Background Analysis, Search Engine-based Image Recognition, and External Tools Application. Additionally, we connect the landmark-level node in the feature set to this node, accounting for the probability that analysts can lock the region by some specific landmarks. Table 2 details the complete connectivity scheme of the Bayesian Network.

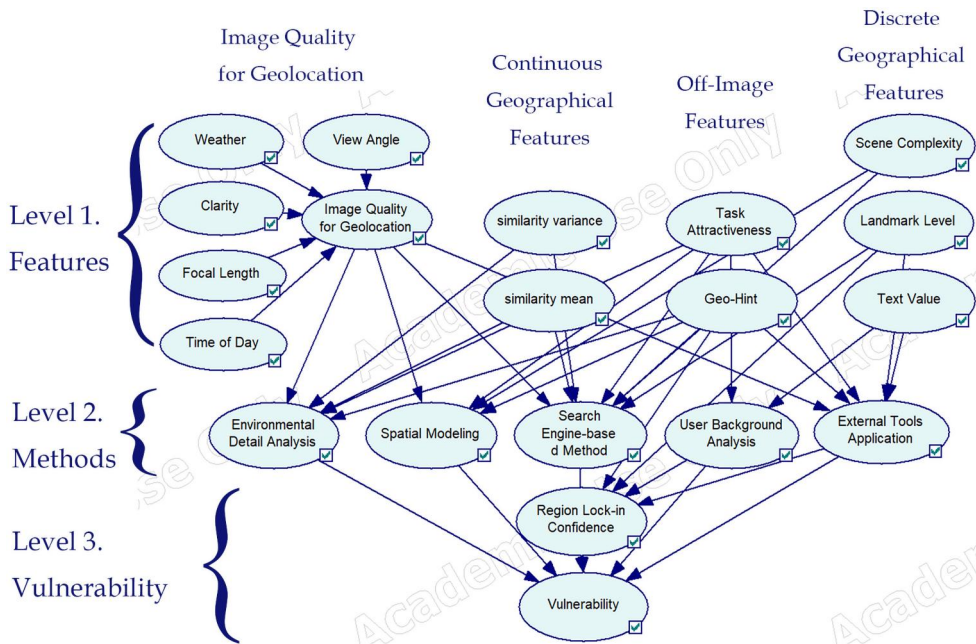


Figure 3. The structure of the hierarchy Bayesian network.

Table 2. Connectivity for intermediate hidden nodes.

Category	Features	Potential localization method				
		Search engine-based method	User background analysis	Spatial modeling	External tool application	Environmental detail analysis
Image quality for geolocation	Image quality for geolocation*	TRUE	FALSE	TRUE	TRUE	TRUE
Continuous geographical features	Similarity variance	TRUE	FALSE	FALSE	FALSE	TRUE
	Similarity mean	TRUE	FALSE	FALSE	FALSE	TRUE
Discrete geographical features	Scene complexity	TRUE	FALSE	TRUE	FALSE	FALSE
	Text value	FALSE	TRUE	FALSE	TRUE	FALSE
	Landmark level	TRUE	TRUE	FALSE	TRUE	FALSE
Off-image features	Geo-Hint	TRUE	TRUE	TRUE	TRUE	TRUE
	TAI Score	TRUE	TRUE	TRUE	TRUE	TRUE

*Image quality for geolocation is associated with image clarity, weather, view angle, focal length, and time of day.

4. Experimental design and results

This section details the experimental design and empirical results. We first describe the curation and characteristics of the ChaoFun42K dataset, followed by detailing the implementation and training procedures for the proposed Bayesian Network. Subsequently, a series of experiments evaluates the model's predictive performance, examines feature sensitivity, and elucidates the reasoning pathways by analyzing the feature influence on geolocation methods.

4.1. Dataset and model implementation

4.1.1. The ChaoFun42K dataset

Existing academic datasets for image geolocation lack two crucial elements for vulnerability analysis: the ground-truth labels for localization success (i.e., whether a case is ‘solved’) and rich, real-world metadata that reflects the OSINT process.

To overcome these limitations, we constructed the ChaoFun42K dataset from ChaoFun², a large, public, Chinese-language OSINT community. Functioning as a crowdsourced challenge platform analogous to r/whereisthis, ChaoFun allows users to post image geolocation challenges for community resolution. We select this community because its challenge-based format provides two crucial elements absent in typical academic datasets: explicit, community-verified labels indicating whether a case is ‘solved’ or ‘unsolved’, and rich, real-world metadata related to user interaction and post context.

To construct this dataset, we systematically collected all publicly available posts from ChaoFun using a custom Python scraper. To ensure privacy, we anonymized all user-identifiable information and filtered out sensitive personal data. Crucially, we extracted the ground-truth label `IsSolved` directly from the post’s metadata tags, assigning `IsSolved=‘TRUE’` only when explicitly marked as ‘Completed’ or ‘Found’ by the community moderators or the original poster. The resulting ChaoFun42K dataset comprises 42,576 anonymized geolocation cases, establishing a robust benchmark for training and validating the proposed risk assessment model. Specifically, the dataset contains 32,889 ‘solved’ and 9,673 ‘unsolved’ cases. Given the community’s high activity level and expertise, we infer that these unsolved cases represent significant geolocation difficulty. [Appendix C](#) details the dataset schema, variable descriptions, and representative examples.

4.1.2. Model training

We randomly partitioned the ChaoFun42K dataset into a training set (80%) and a test set (20%) to assess generalization capability. To prepare the data, we implemented a comprehensive preprocessing and feature extraction pipeline in Python, adhering to the methodological framework detailed in [Section 3](#).

The proposed Bayesian network was implemented and trained in GeNIe software. The implementation process commenced by constructing the network structure according to the specifications in [Section 3](#), ensuring node states correspond to the discretized feature intervals. We then employed the Expectation-Maximization (Dempster *et al.* 1977) algorithm to learn the network parameters (Conditional Probability Tables) from the training data. To mitigate convergence to local optima, we employed random parameter initialization, fixing the random seed at 42 to guarantee reproducibility. [Figure 4](#) visualizes the trained probabilistic structure, including the learned state distributions of nodes.

4.2. Results analysis

For evaluation and presentation, we mapped the model’s continuous vulnerability output to a set of discrete risk intervals via an equal-width binning strategy.

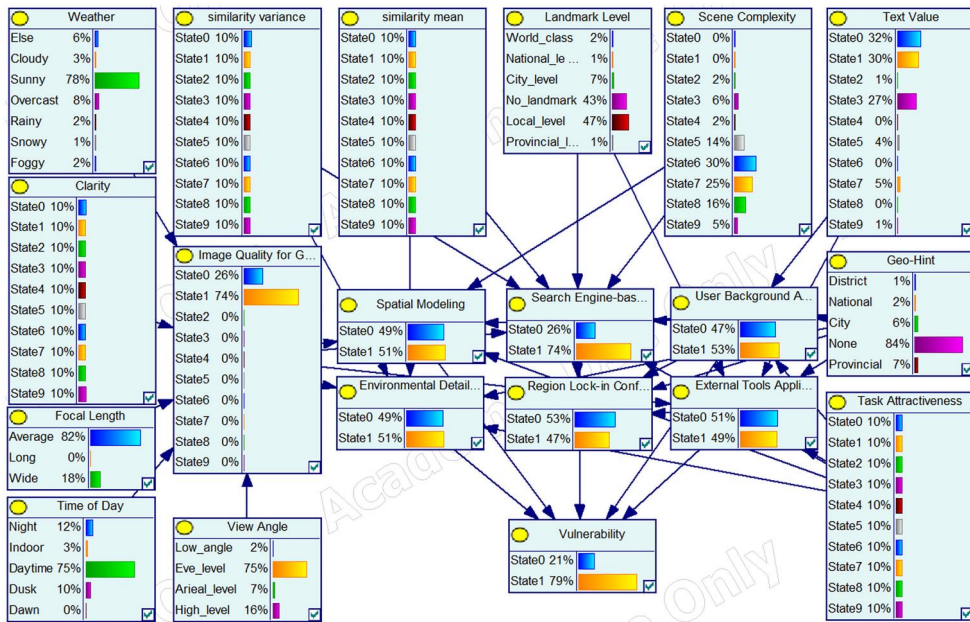


Figure 4. Feature distribution of the trained Bayesian network.

Subsequently, we merged the adjacent pairs of these intervals to form five final risk levels: Super Low (0–0.2), Low (0.2–0.4), Medium (0.4–0.6), High (0.6–0.8), and Super High (0.8–1.0).

4.2.1. Reliability of the vulnerability assessment

To evaluate the model's reliability, we discretized the predicted probabilities for the test set into ten equal-width intervals. For each interval, we defined the 'Ideal Probability' as the midpoint of the interval (e.g., 0.05 for [0, 0.1]). We compared these values with the proportion of solved cases in each interval, derived from the IsSolved ground-truth label. Figure 5(a) visualizes this relationship as a scatter plot, demonstrating a robust correlation between the predicted risk levels and actual outcomes.

According to Figure 5(a), a clear positive correlation is observable. As the predicted ideal probability increases, the actual observed reliability follows a strict linear trend, evidenced by a Pearson correlation coefficient of $r = 0.9951$ ($p < 0.001$). This finding indicates that the model demonstrates strong probabilistic calibration.

To contextualize the model's predictive performance, we benchmarked the proposed model against several standard machine learning classifiers: Logistic Regression, Random Forest, Support Vector Machine (SVM), and XGBoost, all trained on the same feature set.

We adopted the area under the receiver operating characteristic curve (AUC) as the primary evaluation metric. Mathematically, AUC quantifies the probability that the model ranks a randomly selected positive instance higher than a random selected negative one. Given the significant class imbalance (32,889 'solved' vs. 9,673 'unsolved' cases), AUC provides a more robust assessment than accuracy. Figure 5(b) illustrates

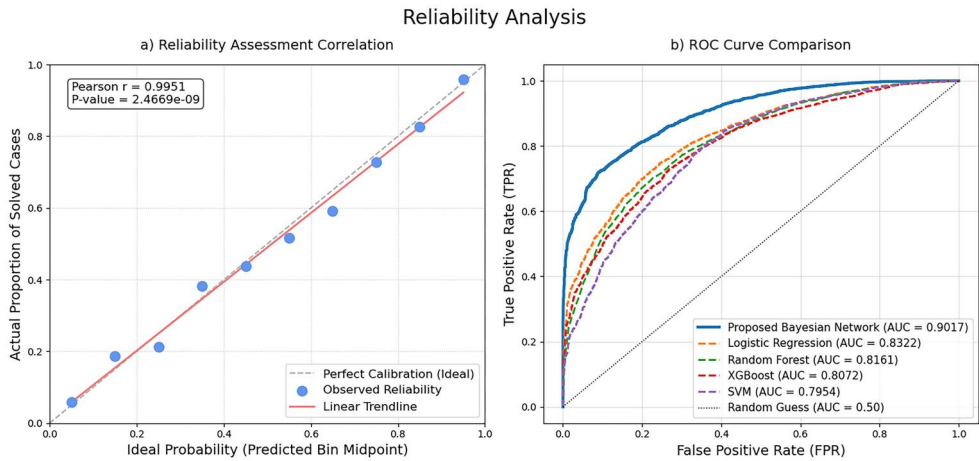


Figure 5. Performance evaluation of the proposed model, displaying (a) the reliability assessment correlation and (b) the ROC curve comparison.

the comparative ROC curves clearly, demonstrating the superior predictive capability of our proposed Bayesian Network.

Figure 6 provides representative case studies, with accompanying bar charts quantifying the contribution of different features to the final vulnerability assessment. This analysis reveals distinct feature attribution patterns across the vulnerability spectrum. Images classified as ‘Super Low’ or ‘Low’ vulnerability typically exhibit minimal discriminative utility. These scenes often depict homogeneous natural landscapes (e.g., sky, open fields), nondescript aerial views, or scenes with limited, unidentifiable details. Consequently, the model correctly identifies these cases as difficult targets, with semantic features such as text value and landmark level showing minimal contribution.

Conversely, images classified as ‘High’ or ‘Super High’ vulnerability are characterized by abundant actionable visual information. They frequently contain prominent landmarks, significant text value, high scene complexity, and strong external indicators (TAI). For instance, the ‘Super High’ vulnerability cases clearly display distinct architectural features and visible signage. The model correctly identifies these features as the primary determinants driving the high-risk assessment.

4.2.2. Sensitivity analysis

4.2.2.1. Feature influence analysis. To systematically quantify the influence of each feature on the final localization probability, we performed two forms of Bayesian network analysis: forward-inference sensitivity analysis and backward-inference diagnostic reasoning. Sensitivity analysis measures the extent to which variations in a single piece of evidence alter the model’s posterior probability. Conversely, diagnostic reasoning facilitates a retrospective approach, identifying the most probable evidence configuration given a known outcome. Figure 7 presented the results of both these dual analyses.

Figure 7(a) illustrates the sensitivity analysis, quantifying the impact of feature state variations on the final probability. We computed a sensitivity index for each feature, defined as the maximum range of $P(\text{IsSolved} = \text{State1})$ observed across all possible

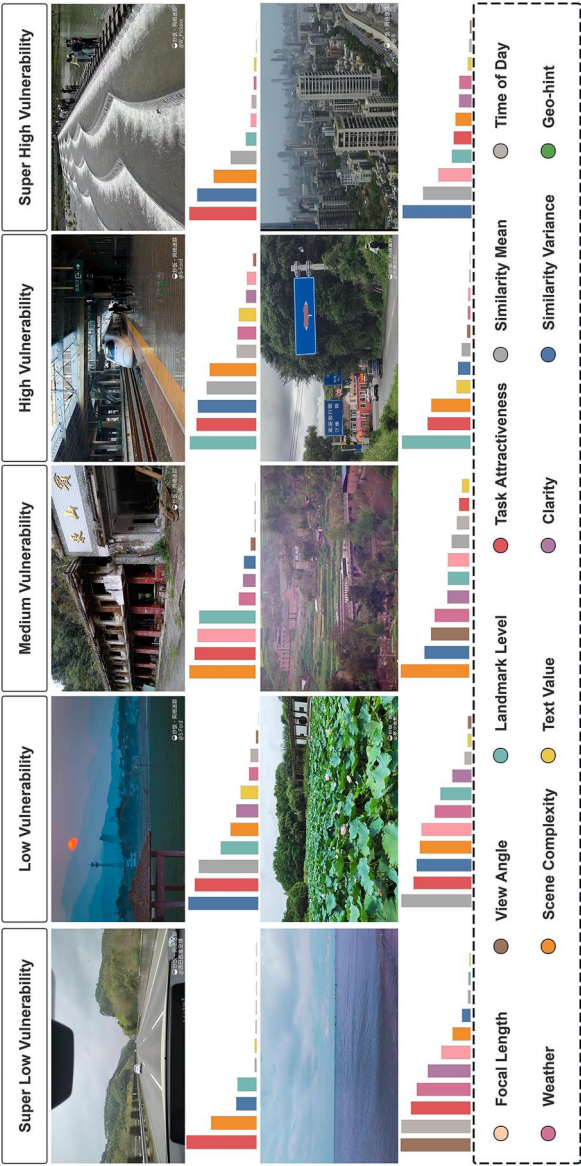


Figure 6. Case studies for photo location disclosure vulnerability analysis.

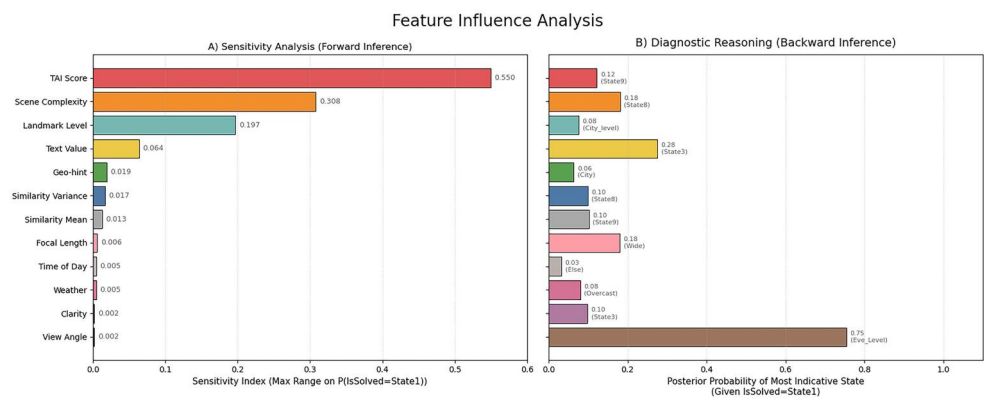


Figure 7. The influence of various factors on the localization result.

feature states. The results identify TAI, scene complexity, and landmark level as the dominant influential factors. The prominence of TAI emphasizes the significance of off-image social context, suggesting that location disclosure risk is often heavily dependent on the publisher’s influence and the post’s timeliness. Concurrently, the high sensitivity of scene complexity and landmark level validates the contribution of discrete geographical features, establishing the presence of unique, identifiable objects as a primary determinant of successful localization.

Figure 7(b) illustrates the diagnostic reasoning analysis, determining the most probable feature configurations conditional on successful localization. We implemented this by setting the evidence `IsSolved=‘True’` and computing the resulting posterior probability distributions of all feature nodes. The bar for each feature shows the posterior probability of its single most indicative state—specifically, the state exhibiting the maximal probability gain compared to its prior distribution. For instance, solved cases exhibit a strong association with high task attractiveness (State9), suggesting that social visibility acts as a key catalyst for collective problem-solving. Similarly, solved images frequently feature high scene complexity(State8) and contain city-level landmarks or geo-hints. These attributes collectively characterize the ‘typical’ profile of a solvable image.

4.2.2.2. Deconstructing localization pathways: feature influence on method selection.

The proposed Bayesian Network explicitly models the intermediate reasoning layer—specifically the five latent localization method nodes. Analyzing the learned probabilistic dependencies between input features (Level 1) and method nodes (Level 2) enables us to deconstruct the reasoning trajectories that an analyst might adopt for a given image.

To achieve this, we conducted a series of one-way sensitivity analyses, targeting each of the five latent method nodes (e.g., `P(Spatial Modeling=‘True’)`). We quantified the sensitivity of each method to every input feature, normalizing the results to represent a relative importance scale. Figure 8 visualizes these relationships via two radar charts, separating features into (a) Structural and Contextual and (b) Photographic and Environmental categories for image geolocation quality. The radar

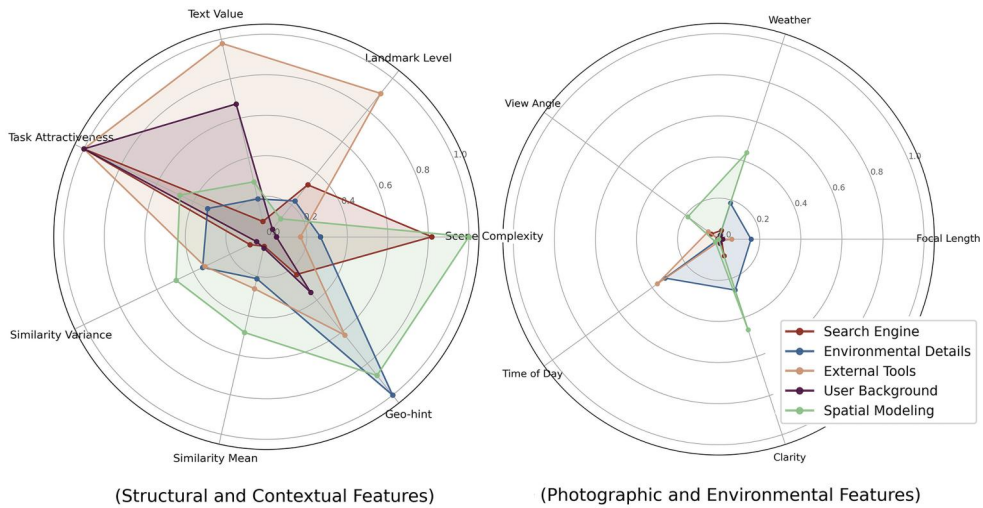


Figure 8. Radar charts illustrating the relative sensitivity of the five localization methods to different input features, revealing the distinct evidence profiles that drive each analytical strategy.

charts illustrate how sensitive each of the five localization methods (colored lines) is to different input features (axes). A point closer to the outer edge (1.0) indicates a stronger dependency.

The analysis in Figure 8 corroborates that distinct localization strategies rely on specific types of evidence. For instance, the Spatial Modeling method (green line) exhibits high sensitivity to scene complexity. This dependency validates the premise that complex structural environments provide the geometric constraints necessary for aligning scene layout with map data. This finding demonstrates the model's capacity to encode the latent reasoning mechanisms of the geolocation process. Consequently, the framework deconstructs the pathways of location disclosure by predicting which analytical methodologies are most likely to be activated by a specific combination of visual and contextual evidence.

5. Conclusion

5.1. Summary of findings and contributions

This study establishes a comprehensive framework for quantitatively assessing geoprivacy risks stemming from implicit geographic cues in social media. By systematically deconstructing visual content and contextual attributes, we constructed a Bayesian Network to model the probabilistic dependencies linking observable features to latent localization methodologies. This approach provides a robust solution for evaluating the inherent geoprivacy risks within User-Generated Content (UGC). Empirical results demonstrate the model's efficacy in discriminating between high-risk and low-risk content ($AUC = 0.90$). Crucially, beyond predictive accuracy, the framework offers an interpretable pathway to understand the underlying mechanisms of geolocation risk formation.

This study advances the geoprivacy paradigm, transcending traditional analyses of explicit ‘point’ (GNSS) and ‘line’ (trajectory) data. We extend the analytical scope to unstructured rich media, conceptually framing visual content as ‘surfaces’ and social context as ‘volumes’. The prominence of the proposed Task Attractiveness Index (TAI) underscores a critical insight: in UGC environments, the scale of collective effort solicited by a post often outweighs intrinsic geospatial content in determining localization vulnerability. This finding suggests that successful localization in open collaborative environments stems not just from a single expert’s analysis, but fundamentally from the complementary nature of the participants’ diverse geographical knowledge.

Furthermore, quantifying feature contributions to localization success yields actionable mitigation strategies for reconciling geoprivacy protection with visual aesthetics. Specifically, we recommend minimizing off-image information leakage and reducing spatially permanent elements within the image to lower its environmental complexity. These strategies effectively disrupt the spatial modeling required for precise localization. Crucially, our findings establish that distinct feature configurations probabilistically trigger specific localization methodologies. This insight provides a foundational basis for the forensic analysis of the image geolocation process, elucidating the latent reasoning processes employed by both human analysts and AI. Ultimately, this research advances a novel theoretical paradigm for digital geoprivacy. By synthesizing social and spatial dimensions, we provide a generalizable methodology applicable to broader GIS risk analysis domains.

5.2. Limitations

Several limitations of this study warrant discussion. Primarily, the generalizability of our findings is constrained by the ChaoFun42K dataset, which, despite its scale, exhibits significant geographic and linguistic biases. Our analysis indicates that over 95% of the cases originate from Chinese-speaking regions. This skew presents a potential source of cultural bias, particularly affecting text-based features like ‘Geo-Hint’ and ‘Text Value’, as norms of information sharing can vary by region.

Compounding this challenge, while the Qwen-VL-Max model demonstrates strong multilingual capabilities, its performance on languages with less representation in its training data—particularly for extracting nuanced, location-specific slang or text from low-quality images—may be limited.

Beyond dataset limitations, other aspects of our methodology present avenues for future refinement. We acknowledge a divergence between our objective risk model and user’s subjective risk perception, which relates to the well-documented ‘privacy paradox’ (Norberg *et al.* 2007). Furthermore, regarding feature formulations, the weighting coefficients within the Task Attractiveness Index (TAI) currently rely on simplifying assumptions. Future research could develop a data-driven approach to optimize these parameters. Similarly, while modeling geolocation methods as discrete latent variables facilitates computational inference, this abstraction simplifies the complex and often fluid reasoning processes of human analysts. Finally, our model predicts a binary ‘solved/unsolved’ outcome, while a more granular risk assessment might consider multiple levels of precision (e.g., country vs. street level) or the time and effort required for localization.

5.3. Practical implications and future research

Our findings have several practical implications. For individual users, this framework lays the groundwork for pre-publication risk auditing tools, enabling users to make risk-aware sharing decisions. For social media platforms, this study underpins the design of algorithmic privacy interventions—specifically, automated risk warnings or intelligent recommendations for content obfuscation. Societally, by quantifying the severity of implicit information leakage, this study provides empirical substantiation for advancing digital privacy literacy education and formulating evidence-based personal information protection regulations.

Finally, from the perspective of AI ethics, this study addresses the ‘black box’ dilemma in GeoAI by offering an interpretable probabilistic model. By illustrating how specific visual and contextual cues trigger localization, our framework enhances algorithmic transparency. This is crucial for mitigating the potential weaponization of open-source data, ensuring that advancements in geospatial intelligence are balanced with robust safeguards against the violation of geoprivacy.

Building upon these findings, several avenues for future research emerge. Future studies could validate the framework on more geographically and culturally diverse datasets to mitigate sampling bias. Concurrently, further studies should quantify the interplay between objective risk and subjective perception, while developing data-driven approaches to optimize the feature parameters. Finally, extending the model to other media formats like video, exploring more granular outcome variables beyond a binary ‘solved/unsolved’ status, and leveraging generative AI for privacy-preserving content modification are also promising directions constitute vital frontiers for the field.

Notes

1. Open-Source Intelligence (OSINT) refers to the collection and analysis of data gathered from public sources available. These sources include the Internet, traditional mass media, academic and public government data, as well as commercial and grey literature. In the context of this paper, OSINT methods are particularly relevant to the process of image geolocation using publicly shared social media data.
2. The specific sub-forum can be accessed at <https://chao.fun/f/84>. As of the time of writing, the community has ceased active geolocation challenges, but the historical case archive remains publicly viewable. Readers can access this dataset at: <https://huggingface.co/datasets/shengjiacheng/Chaofun42K/tree/main>.

Disclosure statement

The authors report there are no competing interests to declare.

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Data and codes availability statement

The data and codes that support the findings of this study are available with the identifier(s) at the DOI link (<https://doi.org/10.6084/m9.figshare.29715128>).

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Appendix A

The prompt used for extracting image features.

Content

ROLE

You are an expert assistant for Open-Source Intelligence (OSINT) and image geolocation (Geoguessing).

Your task is to analyze an image and its accompanying title to extract key features that are crucial for determining the image's geographical location.

INSTRUCTIONS

You will be provided with an image and a title. Analyze both carefully and provide your assessment in a structured JSON format. Adhere strictly to the keys and the possible values provided for each feature.

JSON OUTPUT FORMAT

1. "Geo-Hint":

Description: Assess the level of geographical information provided directly by the post title.
Possible Values: ["None", "National", "Provincial", "City", "District"]

2. "Weather":

Description: Analyze the weather conditions depicted in the image.
Possible Values: ["Sunny", "Cloudy", "Overcast", "Rainy", "Snowy", "Foggy", "Other"]
Note: Use "Other" for indoor scenes or conditions that cannot be determined.

3. "Time of Day":

Description: Estimate the time of day the photo was taken.
Possible Values: ["Daytime", "Night", "Dusk", "Dawn", "Indoor"]
Note: Use "None" for indoor scenes where time of day is not apparent.

4. "View Angle":

Description: Determine the camera's perspective.
Possible Values: ["Eye_level", "High_level", "Low_angle", "Aerial_level"]

5. "Focal Length":

Description: Estimate the lens's field of view (FOV).
Possible Values: ["Average", "Wide", "Long"]

6. "Text Content In Image":

Description: Extract any legible text from the image (e.g., on signs, license plates, advertisements).
Output: An array of strings. If no text is found, return an empty array [].

7. "Text Value":

Description: Based on the extracted text, rate its value for geolocation on a scale of 0–10.
- 0: No text or useless text (e.g., "Welcome").
- 3: Common chain store names (e.g., "McDonald's").
- 7: Highly specific information like a phone number with an area code or a license plate.
- 10: A full, unique address or a highly distinctive and searchable business name.
Output: "State0" to "State9"

8. "Landmark Level":

Description: Identify if the image contains a landmark and assess its level of recognition.
Possible Values: ["No_landmark", "Local_level", "City_level", "Provincial_Level", "National_level", "World_class"]

RESPONSE

Provide ONLY the JSON object in your response, without any additional explanations or markdown formatting.

Example Response:

```
{
  "Geo-Hint": "City",
  "Weather": "Sunny",
  "Time of Day": "Daytime",
  "View Angle": "Eye-level",
  "Focal Length": "Standard",
  "Text Content In Image": ["Main St", "City Bank", "License Plate ABC-123"],
  "Text Value": "State8",
  "Landmark Level": "City_Level"
}
```

Appendix B

This is the list of the selected visual categories for Scene Complexity Calculation. It's a pre-defined list of semantic categories from the ADE20k dataset that are considered to have high informational value for the task of image geolocation. This list excludes elements prone to fragmentation noise (like 'tree' branches) and focuses on stable structures.

Content:

```
KEY_ELEMENT_NAMES = [
  # 1. Buildings and Major Man-made Structures
  'building', 'house', 'skyscraper', 'tower', 'grandstand', 'hovel',
  'wall', 'fence', 'bridge', 'booth', 'stage', 'tent', 'fountain',
  'sculpture', 'pier', 'awning', 'canopy',
  # 2. Infrastructure and Roads
  'road', 'sidewalk', 'path', 'runway', 'dirt track', 'railing', 'bannister',
  'stairs', 'stairway', 'escalator', 'pole', 'streetlight', 'traffic light',
  # 3. Significant Natural Landforms
  'mountain', 'hill', 'rock', 'river', 'waterfall', 'lake',
  # 4. Informative Unique Objects
  'signboard', 'bulletin board', 'flag', 'column'
]
```

Appendix C

Description of the raw data in the ChaoFun42K dataset

Column name	Description	Example value
Title	Post title, which sometimes leaks partial geographic information	The simplest one yet\nChangsha
File Path	Image file path	17227591300.jpg
GmtCreate	Timestamp of post creation	1727700252603
GmtModified	Timestamp of post modification	1727761283241
GmtComment	Timestamp of the last reply to the post	1727710641507
Ups	Number of upvotes for the poster	10
Followers	Number of followers for the poster	1
Tag	Post tag	None / Ancient Problem / Primeval Problem
IsSolved	Solved/unsolved status tag	TRUE / FALSE